Framing, Sentiment, and Agenda-Setting Patterns: A Computational Analysis of News Coverage Across Five Major Outlets

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***Abstract*—This paper presents a large-scale computational study of agenda-setting, sentiment framing, and coverage patterns across five major English-language news outlets—Al Jazeera, CNN, Fox News, PBS NewsHour, and The Guardian—over a seven-month period (July 2025–January 2026). Applying an integrated pipeline of article quality filtering, zero-shot topic classification, transformer-based sentiment analysis, named entity recognition (NER), and TF-IDF lexical profiling to 54,289 articles, we quantify systematic linguistic and structural patterns across publishers. Key findings include: (1) all five outlets allocate 25.2% of total coverage to Politics & Government, yet the ratio of political to science/technology coverage ranges from 9.4:1 to 33.7:1 (11.9:1 overall); (2) Donald Trump receives 112,473 cross-outlet mentions—9.8× more than the second-ranked individual; (3) Politics & Government articles exhibit 44.9% negative sentiment versus 10.2% for Science & Technology; (4) TF-IDF lexical profiling highlights distinct outlet-specific vocabulary patterns; and (5) article-level sentiment is 37% more negative than sentence-level sentiment for Donald Trump, suggesting macro-story context systematically amplifies document-level negativity. These findings provide reproducible cross-outlet empirical benchmarks for computational journalism.**

***Index Terms*—editorial patterns, agenda-setting, sentiment analysis, named entity recognition, computational journalism, framing effects, news analytics.**

I. Introduction

The news media plays a central role in modern democracies, shaping which issues and public figures receive prominent attention and how events are interpreted [1], [2]. Editorial framing may influence public perception through agenda setting, framing, or evaluative tone [3]. Measuring these dimensions across diverse news organizations is both a computational and societal priority.

Existing computational approaches largely examine individual facets of media coverage in isolation, such as political framing and negativity [4], [5] or lexical distinctiveness [6]. Integrated multi-dimensional studies remain rare, and few benchmarks systematically evaluate the interactions between macro-level agenda distributions and micro-level entity portrayals [7]–[9].

To address these limitations, we present an empirical computational investigation of media coverage patterns across 54,289 articles from five major outlets. Our primary methodological and empirical contributions are threefold:

1) **Integrated Multi-Dimensional Framework**: We formulate a multi-tier pipeline unifying zero-shot topic classification, evaluative sentiment scoring, entity salience tracking, and lexical profiling under standardized quality criteria.
2) **Quantification of Aggregation Divergence**: We identify and mathematically operationalize the divergence between whole-article and sentence-level entity sentiment ($\Delta_{\text{negativity}}$), demonstrating a 37% negative intensity distortion induced by macro-document context bleed.
3) **Reproducible Empirical Benchmarks**: We establish cross-outlet baseline metrics for topic crowding-out (11.9:1 politics-to-science ratio), category-level framing asymmetry (4.4× elevation in political negativity), and person-level salience concentration (9.8×).

II. Related Work

The agenda-setting hypothesis [1] posits that mass media influences public priority by allocating disproportionate attention to select issues, often causing systematic topic crowding-out [10]–[12]. Concurrently, evaluative framing is widely operationalized via transformer-based sentiment classification [2], [4], [7], with studies demonstrating pervasive negativity bias in political coverage [5] operating at macro-document and

micro-propositional levels [7], [13]. Furthermore, computational entity-centric analytics leverage NER to quantify figure prominence [14]–[16], while TF-IDF profiling operationalizes linguistic divergence and newspaper slant [6], [13], [17], [18].

While prior computational studies have investigated individual dimensions—such as partisan framing [4], entity visibility [15], or lexical slant [6] in isolation—integrated multi-dimensional studies examining how topic selection, entity salience, evaluative sentiment, and aggregation levels interact across diverse news organizations remain scarce. Furthermore, the methodological distortion induced by whole-article sentiment aggregation versus entity-proximate sentence scoring has not been systematically quantified across large multi-outlet corpora. This study addresses these gaps through an empirical framework covering 54,289 articles.

III. Dataset and Methodology

A. Data Collection and Quality Filtering

News articles were gathered from five major media outlets via public APIs, XML sitemaps, and RSS feeds over a seven-month period (July 2025–January 2026): Al Jazeera, CNN, Fox News, PBS NewsHour, and The Guardian. Editorial orientations follow established AllSides and Media Bias/Fact Check classifications [19], [20], categorized into three analytical cohorts: Center-Left (The Guardian, CNN), Center/International (PBS NewsHour, Al Jazeera), and Right (Fox News; Table I).

To ensure reproducibility, data ingestion followed a standardized multi-stage protocol: (1) *URL Collection and Deduplication*: URLs were canonicalized by stripping tracking query parameters, with duplicate entries pruned via composite key matching on outlet, headline, and publication timestamp. (2) *Full-Text Extraction*: Main-body text was extracted using the high-precision `trafilatura` engine [21], removing navigation headers, advertisements, and comment sections. (3) *Quality Filtering*: A composite quality score $Q \in [0, 1]$ was computed as $Q = 0.7 \times \min(1.0, W/1200) + 0.3 \times \min(1.0, P/20)$, where W is word count and P is paragraph count. Articles with $W < 250$ or $Q < 0.3$ were excluded to eliminate truncated feeds and stub pages.

No artificial downsampling was applied, yielding a complete census of published coverage meeting these quality gates. A total of 54,289 articles passed quality filtering from an initial corpus of 56,256 collected entries (Table I and Fig. 1).

TABLE I
Dataset Summary and Political Orientation by News Outlet

Outlet	Orientation [19]	Collected	Retained	Share
The Guardian	Center-Left	31,206	29,784	54.8%
CNN	Center-Left	9,504	9,186	16.9%
Al Jazeera	Center / Int.	7,097	7,073	13.0%
Fox News	Right	4,704	4,502	8.3%
PBS NewsHour	Center	3,745	3,744	6.9%
Total	—	**56,256**	**54,289**	**100.0%**

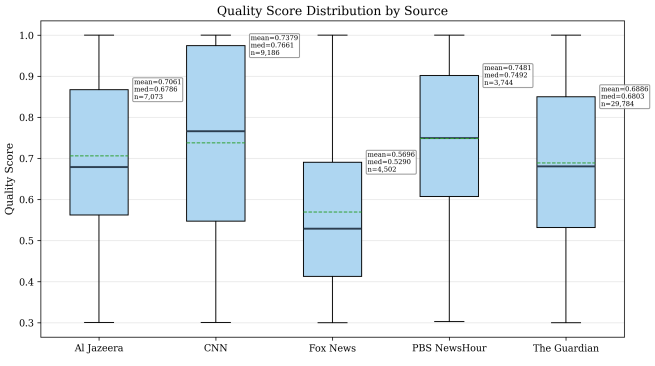


Fig. 1. Article quality score distributions by outlet.

B. Topic Classification

A zero-shot Natural Language Inference (NLI) model, `facebook/bart-large-mnli` [22]–[24], was applied to classify each article into one of 11 defined categories: Business & Economy, Crime & Justice, Education, Entertainment & Culture, Environment & Climate, Health & Medicine, International Affairs & Diplomacy, Military & Defense, Politics & Government, Science & Technology, and Sports. Formulating classification via the template *"This article is about {}."* enables standardized, scalable categorization across diverse news outlets. Long articles were partitioned into overlapping token chunks (600 tokens per chunk with 100-token overlap) to prevent boundary truncation. The model output achieved a mean Top-1 confidence score of 0.873 (Fig. 2, right), with only 0.4% of articles ($N = 214$) falling below 0.40 confidence. This high confidence distribution reflects consistent model classification certainty.

C. Sentiment Analysis

Sentiment polarity was quantified using the fine-tuned RoBERTa-large classifier `j-hartmann/sentiment-roberta-large-english-3-classes` [25], [26], trained on multi-domain sentiment benchmarks across positive, neutral, and negative classes. The model outputs posterior probabilities: $P(\text{positive})$, $P(\text{neutral})$, and $P(\text{negative})$, where discrete sentiment is assigned via $\arg\max_c P(c)$. A continuous compound polarity score $S \in [-1.0, +1.0]$ was defined as $S = P(\text{positive}) - P(\text{negative})$. For article-level sentiment, texts exceeding the 512-token limit were split into sliding-window chunks (400 tokens with 50-token overlap) and aggregated via unweighted mean pooling across chunk probabilities. For sentence-level analysis, individual sentences referencing named figures were extracted using the transformer-based spaCy `en_core_web_trf` pipeline [16] and scored independently.

D. Named Entity Recognition and Lexical Profiling

Named entity recognition (NER) was conducted using the transformer-based spaCy pipeline [16] with coreference resolution to unify name variants into canonical entities. Mentions were disaggregated across outlet, topic category, and month

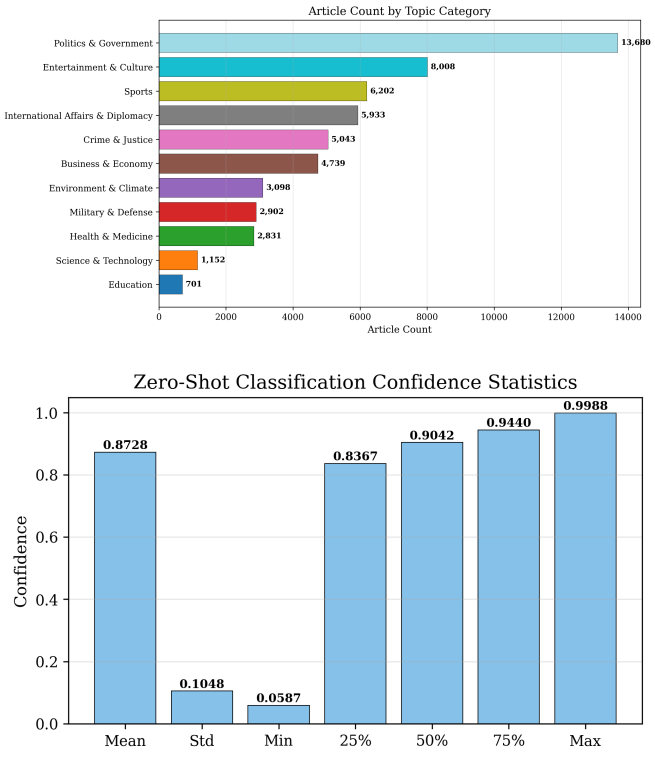


Fig. 2. Topic distribution (top) and zero-shot classification confidence (bottom).

for multi-dimensional salience tracking. Outlet-specific lexical distinctiveness was quantified using TF-IDF feature extraction ($2 \leq \text{df} \leq 0.90 \times |D|$) with standard stopword pruning, computing mean feature weights $\bar{w}_{t,s}$ per outlet sub-corpus to extract characteristic vocabulary signatures.

IV. Results: Dimensions Of Editorial Divergence

A. Topic-Selection Patterns

Fig. 2 illustrates the empirical distribution across the 11 topic categories. Politics & Government received the largest share of coverage with $N = 13,680$ articles (25.2% of the corpus), whereas Science & Technology received $N = 1,152$ (2.1%), and Education received $N = 701$ (1.3%). Across the entire corpus, this yields an overall ratio of 11.9:1 between political and science/technology coverage.

Disaggregating across outlets (Fig. 3) reveals substantial variation in topic allocation: Al Jazeera exhibits a political-to-science ratio of 33.7:1 (1,718:51), Fox News 16.2:1 (1,397:86), PBS NewsHour 16.0:1 (1,152:72), The Guardian 10.2:1 (6,731:659), and CNN 9.4:1 (2,682:284). These differences illustrate measurable variation in editorial emphasis across newsrooms.

Regarding genre-specific overrepresentation, The Guardian generated the majority of Entertainment & Culture coverage (6,337 articles; 79.1% of the corpus total) and Sports (4,361; 70.3%), aligning with its broad general-interest scope. Conversely, Fox News showed high relative focus on Crime & Justice (741 articles; 16.5% of its coverage vs. 9.3% corpus average). Al Jazeera dedicated 32.6% of its total output ($N = 2,304$) to International Affairs & Diplomacy compared to the 10.9% corpus average (Fig. 3).

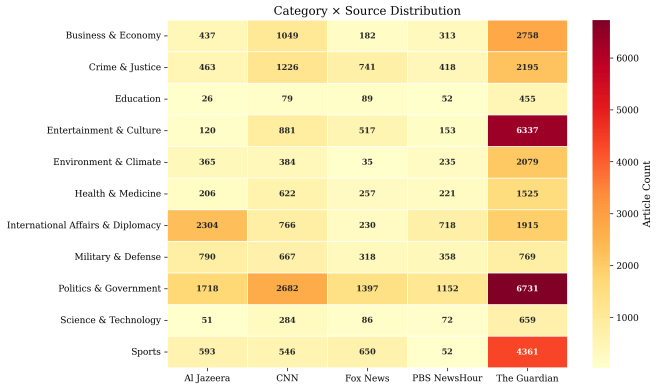


Fig. 3. Topic–source article count heatmap.

Temporal analysis (Fig. 4) indicates that publication volume peaked in December 2025, driven by Politics & Government ($N = 2,731$) and Sports ($N = 1,298$). Total monthly volume exceeded 10,800 articles across all topics in December 2025, representing a 45% increase over the July 2025 baseline.

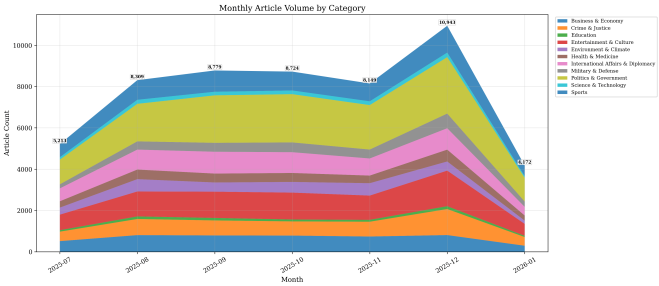


Fig. 4. Monthly article volume by topic category.

B. Sentiment and Framing Patterns

All five outlets exhibited negative mean compound sentiment across full article texts, with categorical negative shares spanning 18.7% (PBS) to 31.7% (Guardian) (Fig. 5) [5]. Aggregating by political leaning shows comparable negative baselines across Center (−0.26), Center-Left (−0.29), and Right (−0.29) cohorts, demonstrating pervasive ideological negativity.

Category-level framing (Fig. 6) shows Politics & Government carries the highest proportion of negative coverage (44.9%), 4.4× greater than Science & Technology (10.2%). Crime & Justice (38.1%) and Education (37.2%) also display heavy negative framing, while Sports (14.9%) and Entertainment & Culture (20.8%) remain predominantly neutral or positive.

The title–content sentiment gap (Fig. 7) exhibits a tight distribution centered near zero (mean ≈ 0), indicating close

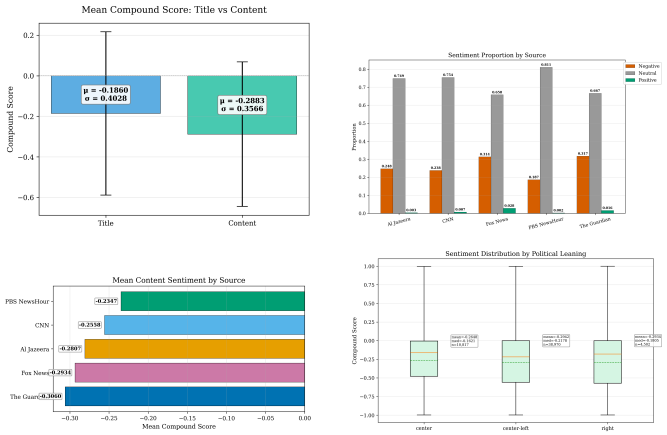


Fig. 5. Sentiment overview: compound score distributions (top-left), content sentiment proportions by outlet (top-right), mean scores by outlet (bottom-left), and sentiment by political leaning (bottom-right).

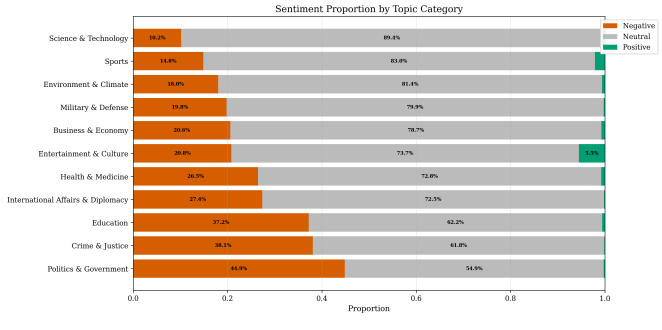


Fig. 6. Sentiment proportions by topic category.

empirical alignment between headline framing and full-text sentiment.

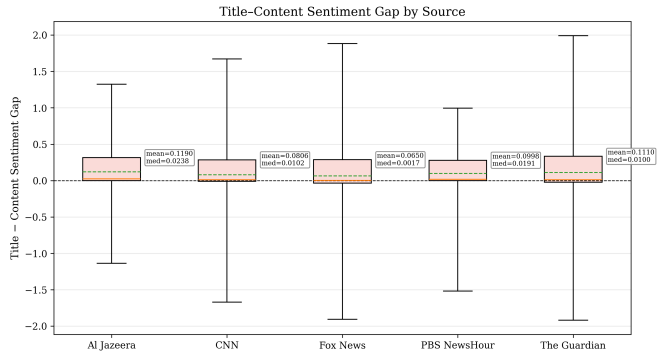


Fig. 7. Distribution of title–content sentiment gaps.

C. Named Entity Salience Patterns

Donald Trump received the highest volume of mentions ($N = 112,473$) across the corpus—9.8× more than the second-ranked individual (Jeffrey Epstein; 11,511) and 12.9× more than the third-ranked (Vladimir Putin; 8,738). The full top-30 ranking is presented in Fig. 8. A chi-squared proportional coverage test against an expected distribution based on baseline publication volume confirms that inter-outlet mention allocation is significantly non-uniform ($\chi^2 = 31{,}351.3$, $p < 0.001$), indicating pronounced salience concentration on a single political figure across all outlets.

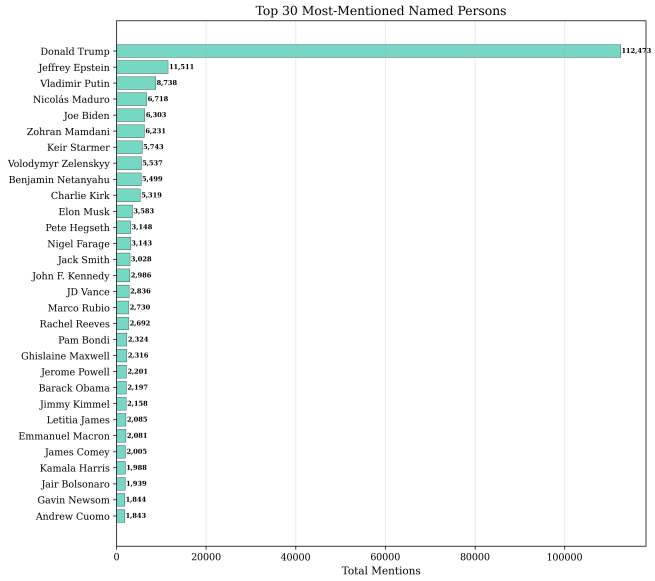


Fig. 8. Top 30 most-mentioned named persons.

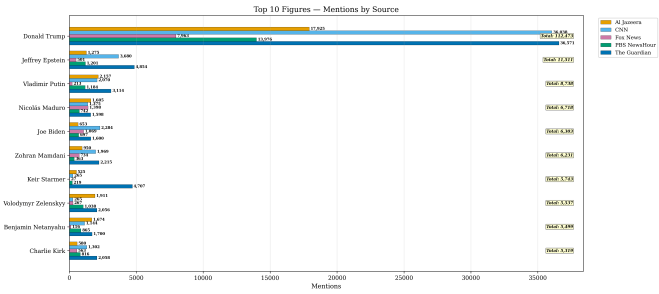


Fig. 9. Mention counts of top 10 figures across outlets.

Disaggregating Trump mentions by outlet (Fig. 9) shows that CNN and The Guardian published the highest raw volumes. Normalizing by retained publication volume reveals distinct entity density: CNN recorded 3.92 mentions/article and PBS NewsHour 3.73, followed by Al Jazeera (2.53), Fox News (1.77), and The Guardian (1.23; corpus average: 2.07), showing higher figure concentration in dedicated broadcast reporting. Grouping by political orientation (Fig. 10), center-left outlets accounted for 64.6% of total Trump mentions, center/international outlets 28.4%, and right-leaning Fox News 7.1%.

Monthly trajectory analysis (Fig. 11) indicates that Trump coverage maintained a sustained baseline of 12,109 to 19,632 monthly mentions between July 2025 and January 2026, peaking in December 2025. In contrast, other figures exhibit episodic peaks, such as Jeffrey Epstein in November 2025

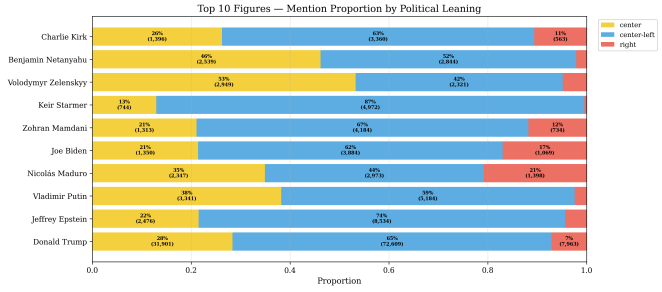


Fig. 10. Mention distributions of top figures by political leaning.

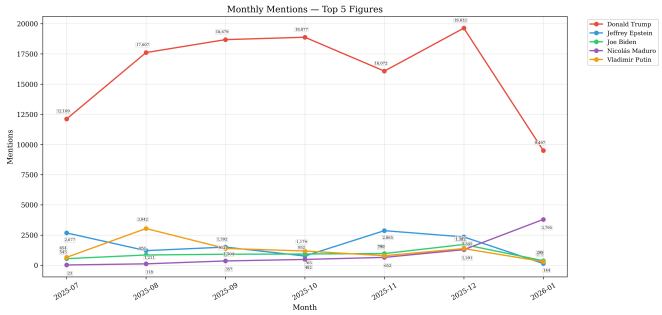


Fig. 11. Monthly mention trends for top five figures.

($N = 2,800$+) and Nicolás Maduro in January 2026 ($N = 3,795$), demonstrating the contrast between persistent figure salience and event-driven episodic attention.

D. Sentiment Variations Toward Named Figures

Article-level sentiment scores for the top 10 mentioned figures (Fig. 12) are consistently negative across all individuals, ranging from −0.20 (Zelenskyy) to −0.48 (Epstein). None of the top 10 figures received net positive or net neutral coverage in aggregate.

To evaluate whether whole-article sentiment accurately reflects individual portrayal, we compared article-level compound scores with isolated sentence-level compound scores for sentences containing each entity mention (Fig. 12). The aggregation divergence in negative tone is quantified as $\Delta_{\text{negativity}} = (|\bar{S}_{\text{article}}| - |\bar{S}_{\text{sentence}}|)/|\bar{S}_{\text{sentence}}| \times 100\%$. For Donald Trump ($N = 107,640$ sentence instances), the article-level mean compound score is −0.397 compared to a sentence-level mean of −0.290, yielding a 36.9% (≈ 37%) relative increase in negative intensity at the article level (Spearman $\rho = 0.348$, $p < 0.001$). This systematic gap suggests that macro-level story context (e.g., controversies, legislative battles, geopolitical conflicts) may influence article-level figure evaluations, potentially inflating perceived personal negativity compared to local sentence framing. Cross-tabulating discrete sentiment labels at both granularities (Fig. 13) confirms that 64.8% of articles classified as negative at the document level contain predominantly neutral sentences, quantifying the aggregation artifact.

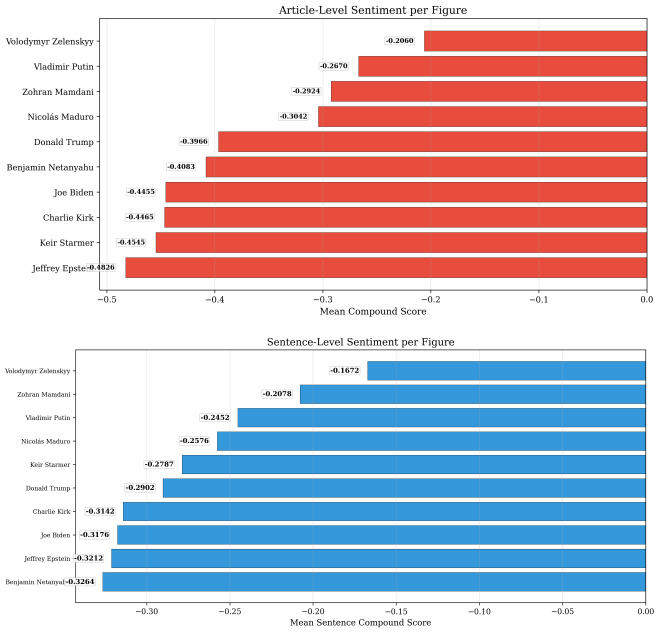


Fig. 12. Comparison of article-level (top) and sentence-level (bottom) sentiment per named figure. Article-level scores are systematically more extreme.

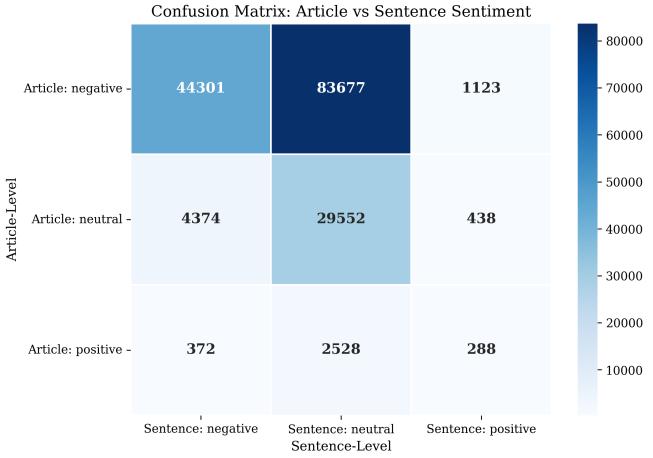


Fig. 13. Article vs. sentence-level sentiment confusion matrix.

E. Lexical Profiling and Vocabulary Selection

TF-IDF feature extraction reveals distinct outlet-specific lexical signatures aligned with editorial mandates: Al Jazeera emphasizes international conflict terminology (*israel, gaza, military, prime minister, united nations*), Fox News highlights digital reporting and domestic enforcement (*news digital, president donald, law enforcement*), The Guardian reflects UK governance and global affairs (*uk, world, prime minister, human rights, far right*), and CNN and PBS prioritize federal institutions (*white house, federal, administration, justice department, supreme court*), confirming systematic vocabulary differentiation across publishers.

V. Discussion

Despite contrasting orientations, all publishers share a thematic hierarchy: Politics & Government (25.2%), Entertainment & Culture (14.8%), and Sports (11.4%). The 9.8× Trump salience concentration reflects heightened executive newsworthiness during an active election cycle rather than isolated editorial slant [27]. While slant appears in framing and vocabulary, thematic convergence is driven by news production dynamics: commercial attention economies [28], 24-hour cycle demands, and intermedia agenda mimicry [10], [11]. Outlet-level specialization—such as Al Jazeera's international focus or Fox's crime reporting—reflects distinct organizational mandates.

Our findings offer two key methodological takeaways for NLP-driven media analytics. First, the 37% aggregation divergence reveals that whole-article sentiment is an unreliable proxy for entity-level evaluation; macro-level narrative context appears to influence figure-level scores. Future computational studies must adopt target-proximate or syntax-aware sentence scoring to avoid this ecological distortion. Second, media literacy and automated auditing tools must evaluate multi-dimensional coverage profiles—incorporating topic crowding and entity salience—alongside conventional sentiment polarity metrics [27]. The study's major limitations include: (1) coverage is restricted to five publishers; (2) quality score thresholding ($Q \geq 0.3$) filters out short-form articles; (3) zero-shot classification, despite achieving a high mean confidence of 0.873, reflects model inference certainty rather than human-validated ground truth, motivating future multi-annotator validation benchmarks; (4) the seven-month collection timeframe (July 2025–January 2026) captures a specific election cycle; (5) political leaning labels are derived from external aggregators (AllSides, MBFC) [19], [20]; and (6) automated coreference resolution may introduce minor entity aggregation noise [29].

VI. Conclusion

This study presented a computational analysis of editorial patterns across five news publishers. Key findings include: an 11.9:1 politics-to-science coverage ratio; a 9.8× salience concentration around one figure; a 4.4× elevation in negative framing for political topics; and a 37% overestimation of person-level negativity under whole-article sentiment extraction relative to sentence-level scoring. Future work will extend this framework across multilingual corpora, extended timeframes, and dynamic network agenda-setting models [30].

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